

$$\begin{aligned}
 13.1 \quad \int_{\partial \Omega} v \frac{\partial u}{\partial n} d\sigma &= \int_{\partial \Omega} v \nabla u \cdot n d\sigma = \int_{\partial \Omega} v \nabla u \cdot n d\sigma \\
 &= \int_{\Omega} \Delta (v \nabla u) dV \\
 &= \int_{\Omega} (v \Delta u + \nabla u \cdot \nabla v) dV \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 \text{利用 (1) 式, 从而有 } \int_{\partial \Omega} (u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n}) d\sigma &= \int_{\Omega} (u \Delta v + \nabla u \cdot \nabla v - v \Delta u - \nabla v \cdot \nabla u) dV \\
 &= \int_{\Omega} (u \Delta v - v \Delta u) dV
 \end{aligned}$$

2. 必要性: $\int_{\partial \Omega} \frac{\partial u}{\partial n} d\sigma = \int_{\partial \Omega} \nabla u \cdot n d\sigma = \int_{\text{int } \Omega} \Delta u dV = 0$

充分性: 若 $\exists x_0 \in D$ 使 $\Delta u(x_0) \neq 0$

由于 $u \in C^2(D, \mathbb{R})$, 从而有存在 $\delta > 0$, 使 $\forall x \in B(x_0, \delta)$, $|\Delta u(x) - \Delta u(x_0)| < |\Delta u(x_0)|$

不妨设 $\Delta u(x_0) > 0$ 则 $\Delta u(x) > 0$

$$\therefore \text{取 } C = \partial B(x_0, \delta) \text{ 则 } \int_C \frac{\partial u}{\partial n} d\sigma = \int_{B(x_0, \delta)} \Delta u dV > 0$$

矛盾故 u 在 D 内为调和函数

$$3. (1)(a) \text{ 有 } \frac{\partial}{\partial y} \left(\frac{x}{1+x^2+y^2} \right) = \frac{\partial}{\partial x} \left(\frac{y}{1+x^2+y^2} \right)$$

设左边为 $\frac{\partial P}{\partial y}$, 右边为 $\frac{\partial Q}{\partial x}$, 则

$$\frac{\partial P}{\partial x} = \frac{\partial P}{\partial y} = \frac{-2xy}{(1+x^2+y^2)^2} \quad \frac{\partial Q}{\partial x} = \frac{-y^2 f'(x)}{(1+x^2+y^2)^2}$$

从而有 $f'(x) = 2x$ 又有 $f(0) = 1$, 故 $f(x) = x^2 + 1$

(b) 由于该函数积分路径与半径无关

(b) 该函数积分与路径无关, 故直接选取直线 $y=0$

$$\text{从而 } I = \int y \frac{xdx + ydy}{x^2 + y^2} = \int_0^2 \frac{x dx}{x^2 + 1} = \frac{1}{2} \int_0^4 \frac{dx}{x+1} = \frac{1}{2} \ln 5$$

(2) 设 $P = (e^x + f(x))y$ $Q = -f(x)$

$$\text{则有 } \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \text{ 即 } e^x + f(x) = -f'(x)$$

$$\text{得 } f(x) = (x - \frac{1}{2})e^x - \frac{1}{2}e^x$$

$$\text{路径取 } y=x, \text{ 有 } \int_{(0,0)}^{(1,1)} (e^x + f(x))y dx - f(x)dy = \int_0^1 e^x dx = e - 1$$

$$4. \text{ 路径取 } y=0 \text{ 和 } x=1, \text{ 有 } \int_{(0,0)}^{(1,1)} (e^x + f(x))y dx - f(x)dy = \int_0^1 -f(x)dx + \int_1^1 -f(x)dy = \frac{e}{2}$$

$$4. \text{ 计算 } \frac{\partial F'_i(x)}{\partial x_j} = \frac{\partial f(x_i)x_i}{\partial x_j} = x_i \frac{\partial f(x_i)}{\partial x_j} = f'(x_i) \frac{x_i x_j}{|x_i|}$$

$$\therefore \frac{\partial F'_i(x)}{\partial x_j} = \frac{\partial F'_j(x)}{\partial x_i} \text{ 即有 } F(x) \text{ 是对称阵, } F \text{ 是无旋场, 从而 } F \text{ 是势场}$$

$$5. \nabla f(x) = \nabla g(|x|) = g'(|x|) \frac{x}{|x|}$$

$$\Delta f(x) = g''(|x|) \frac{x \cdot x}{|x|^2} + \frac{|x|}{|x|^2} g'(|x|)$$

$$5. \nabla f(x) = \nabla g(|x|) = \sum_{i=1}^n g'(|x|) \frac{x_i}{|x|}$$

$$\Delta f(x) = \sum_{i=1}^n \nabla \cdot (g'(|x|) \frac{x_i}{|x|}) = \sum_{i=1}^n (g''(|x|) \frac{x_i^2}{|x|^2} + g'(|x|) (\frac{1}{|x|} - \frac{x_i^2}{|x|^3})) = g''(|x|) + \frac{n-1}{|x|} g'(|x|)$$

$$\text{令 } \Delta f(x) = 0, \text{ 从而 } g''(|x|) + \frac{n-1}{|x|} g'(|x|) = 0$$

$$\therefore g'(|x|) = A |x|^{-(n-1)}$$

$$n=2 \text{ 时 } g(x) = C_1 \ln x + C_2$$

$$n \neq 2 \text{ 时 } g(x) = C_1 x^{2-n} + C_2$$

$$\therefore f(x) = \begin{cases} C_1 \ln x + C_2 & n=2 \\ C_1 x^{2-n} + C_2 & n \neq 2 \end{cases}$$

6. 设 u 的势函数为 v 即 $\nabla v = u$

根据不可压流体的定义 $\nabla \cdot u = 0$

$$\therefore \Delta v = 0$$

即 v 是调和函数

$$7. (1) \nabla \cdot (aA + bB) = \sum_{i=1}^n \frac{\partial (aA + bB)}{\partial x^i} = a \sum_{i=1}^n \frac{\partial A}{\partial x^i} + b \sum_{i=1}^n \frac{\partial B}{\partial x^i} = a \nabla \cdot A + b \nabla \cdot B$$

$$\nabla(aA + bB) = \sum_{i=1}^n \frac{\partial (aA + bB)}{\partial x^i} x^i = a \sum_{i=1}^n \frac{\partial A}{\partial x^i} x^i + b \sum_{i=1}^n \frac{\partial B}{\partial x^i} x^i = a \nabla A + b \nabla B$$

$$\begin{aligned} \nabla \times (aA + bB) &= \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ aA_1 + bB_1 & aA_2 + bB_2 & aA_3 + bB_3 \end{vmatrix} = (aA_1' + bB_1') \begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{vmatrix} + (aA_2' + bB_2') \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial z} \end{vmatrix} + (aA_3' + bB_3') \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \end{vmatrix} \\ &= a \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix} + b \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ B_1 & B_2 & B_3 \end{vmatrix} = a \nabla \times A + b \nabla \times B \end{aligned}$$

$$\Delta(aA + bB) = \nabla \cdot \nabla(aA + bB) = \nabla \cdot (a \nabla A + b \nabla B) = a \nabla \cdot \nabla A + b \nabla \cdot \nabla B = a \Delta A + b \Delta B$$

$$(2) \nabla \cdot (fA) = \sum_{i=1}^n \frac{\partial (fA_i)}{\partial x^i} =$$

$$(2) \nabla \cdot (fA) = \sum_{i=1}^n \frac{\partial (fA_i)}{\partial x^i} = \sum_{i=1}^n \frac{\partial f}{\partial x^i} A_i + \sum_{i=1}^n f \frac{\partial A_i}{\partial x^i} = (\nabla f) \cdot A + f \nabla \cdot A$$

$$\begin{aligned} \nabla \times (fA) &= \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ fA_1 & fA_2 & fA_3 \end{vmatrix} = fA_1 \begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{vmatrix} (fA) + \begin{vmatrix} \frac{\partial}{\partial z} & \frac{\partial}{\partial x} \end{vmatrix} (fA_2) + \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \end{vmatrix} (fA_3) \\ &= \begin{vmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix} + f \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix} \\ &= (\nabla f) \times A + f \nabla \times A \end{aligned}$$

$$(3) \nabla \cdot \nabla f = \nabla \cdot \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = \sum_{i=1}^3 \frac{\partial^2 f}{\partial x^i \partial x^i} = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = \Delta f$$

$$\nabla \times \nabla f = \nabla \times \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix} = 0$$

$$\begin{aligned} (4) \nabla \cdot (\nabla \times A) &= \nabla \cdot \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z}, \frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x}, \frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \\ &= \frac{\partial^2 A_3}{\partial y \partial x} - \frac{\partial^2 A_2}{\partial z \partial x} + \frac{\partial^2 A_3}{\partial z \partial y} - \frac{\partial^2 A_3}{\partial x \partial y} + \frac{\partial^2 A_2}{\partial x \partial z} - \frac{\partial^2 A_1}{\partial y \partial z} \\ &= 0 \end{aligned}$$

8. 由 (4) $\nabla \cdot G = \nabla \cdot \text{rot } F = 0$ 设 $f(x) = \lambda^3 x$

$$\therefore \nabla \cdot g f = (\nabla g) f + g \nabla \cdot f = (\nabla g) f + 3g = 0$$

$$\text{设 } h(\lambda) = \lambda^3 g(\lambda p) \quad \therefore h'(\lambda) = \frac{d}{d\lambda} \lambda^3 g(\lambda p) + \lambda^3 \nabla g(\lambda p) p = \lambda^2 (3g(\lambda p) + \nabla g(\lambda p)(\lambda p)) = 0$$

$$\therefore h(\lambda) \text{ 是常数 } C \quad \text{若 } C \neq 0 \quad \therefore h(\lambda) = C = h(0) = 0 \quad \therefore g(\lambda p) = 0 \quad \text{即 } g \equiv 0$$

9. (1) 取 $F_1 = \nabla \varphi$ $F_2 = F - \nabla \varphi$

则 $\nabla \times F_1 = \nabla \times \nabla \varphi = 0$ $\nabla \cdot F_2 = \nabla \cdot (F - \nabla \varphi) = \nabla \cdot F - \Delta \varphi = 0$

(2) 根据场论基本定理, 若 F_1 在 Ω 上无旋, 则存在 $\nabla \varphi = F_1$.

从而 $F_2 = F - F_1 = F - \nabla \varphi$

$\therefore \nabla \cdot F_2 = \nabla \cdot F - \Delta \varphi \quad \therefore \Delta \varphi = \nabla \cdot F$

10. f) $\nabla \times (A \times B) = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 B_1 - A_3 B_3 & A_3 B_1 - A_1 B_3 & A_1 B_2 - A_2 B_1 \end{vmatrix} = (B \cdot \nabla) A + (\nabla \cdot B) A - (\nabla \cdot A) B - (A \cdot \nabla) B$

h) $\nabla \cdot (A \times B) = \nabla \cdot (A_2 B_3 - A_3 B_2, A_3 B_1 - A_1 B_3, A_1 B_2 - A_2 B_1)$
 $= \frac{\partial}{\partial x_1} (A_2 B_3 - A_3 B_2) + \frac{\partial}{\partial x_2} (A_3 B_1 - A_1 B_3) + \frac{\partial}{\partial x_3} (A_1 B_2 - A_2 B_1)$
 $= \frac{\partial A_2 B_3}{\partial x_1} + \frac{\partial A_2 \partial B_3}{\partial x_1} - \frac{\partial B_3 A_2}{\partial x_1} - \frac{A_3 \partial B_2}{\partial x_1} + \frac{\partial B_2 A_3}{\partial x_2} + \frac{A_3 \partial B_1}{\partial x_2} - \frac{A_1 \partial B_3}{\partial x_2} - \frac{B_2 \partial A_1}{\partial x_2} + \frac{B_2 \partial A_1}{\partial x_3} + \frac{A_1 \partial B_2}{\partial x_3} - \frac{B_1 \partial A_2}{\partial x_3} - \frac{A_2 \partial B_1}{\partial x_3}$
 $= B \cdot (\nabla \times A) - A \cdot (\nabla \times B)$

11. a) $\nabla \times \nabla f = 0$ $\nabla \cdot (\nabla \times A) = 0$ $\nabla \nabla \cdot A$ $\nabla \times (\nabla \times A)$ $\nabla \cdot \nabla f$

b) 在 13.7 中已验证

c) $\nabla \times (\nabla \times A) = \nabla (\nabla \cdot A) - \Delta A$

$\nabla \times A = (\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z}, \frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x}, \frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y}) = (B_1, B_2, B_3)$

$\nabla \times (\nabla \times A) = i(\frac{\partial B_3}{\partial y} - \frac{\partial B_2}{\partial z}) + j(\frac{\partial A_1}{\partial z} - \frac{\partial B_3}{\partial x}) + k(\frac{\partial B_2}{\partial x} - \frac{\partial B_1}{\partial y})$

经验证右端

第一方向分量为 $i(\frac{\partial^2 A_3}{\partial x \partial y} - \frac{\partial^2 A_2}{\partial y^2} - \frac{\partial^2 A_3}{\partial z^2} + \frac{\partial^2 A_2}{\partial x \partial z})$

对 $\nabla (\nabla \cdot A) - \Delta A$ 的 x 分量上为

$\frac{\partial}{\partial x} (\nabla \cdot A) - \Delta A_x = \frac{\partial^2 A_x}{\partial x^2} + \frac{\partial^2 A_y}{\partial x \partial y} + \frac{\partial^2 A_z}{\partial x \partial z} - \frac{\partial^2 A_x}{\partial x^2} - \frac{\partial^2 A_y}{\partial y^2} - \frac{\partial^2 A_x}{\partial z^2}$
 $= \frac{\partial^2 A_y}{\partial x \partial y} + \frac{\partial^2 A_z}{\partial x \partial z} - \frac{\partial^2 A_x}{\partial y^2} - \frac{\partial^2 A_x}{\partial z^2}$

从而两者 x 分量一致, y, z 同理

d) 有 $\nabla \times (\nabla \times A) = \nabla (\nabla \cdot A) - \Delta A$

向量公式有 $A \times (B \times C) = B(A \cdot C) - C(A \cdot B)$

$\therefore \nabla \times (\nabla \times A) = \nabla (\nabla \cdot A) - A \nabla^2 = \nabla (\nabla \cdot A) - \nabla^2 A$

12. $\nabla \cdot E = \frac{\rho}{\epsilon_0}$ $\frac{\partial}{\partial t} \nabla \cdot E = \frac{1}{\epsilon_0} \frac{\partial \rho}{\partial t} = \nabla \cdot (\frac{\partial E}{\partial t})$

而对 $\nabla \times B = \frac{1}{\epsilon_0 c^2} j + \frac{1}{c^2} \frac{\partial E}{\partial t}$ 有

$0 = \nabla \cdot (\nabla \times B) = \frac{1}{\epsilon_0 c^2} \nabla \cdot j + \frac{1}{c^2} \nabla \cdot \frac{\partial E}{\partial t} = \frac{1}{\epsilon_0 c^2} \nabla \cdot j + \frac{1}{c^2} \frac{1}{\epsilon_0} \frac{\partial \rho}{\partial t}$

$\therefore \nabla \cdot j = - \frac{\partial \rho}{\partial t}$

13. a) 在球内, 电场为 0

球外, 取 ~~距离球心~~ 半径 R , 球心同该球面的球面, 有

$$\int_S \nabla \cdot \mathbf{E} d\sigma = 4\pi R^2 E = \frac{Q}{\epsilon_0} \quad E = \frac{Q}{4\pi R^2 \epsilon_0}$$

故 $\mathbf{E} = \begin{cases} 0, & R \leq r \\ \frac{Q}{4\pi R^2 \epsilon_0}, & R > r \end{cases}$

点电荷产生的电场为 $E = \frac{q}{4\pi r^2 \epsilon_0}$

二者相同

14. e) 设这两个调和函数为 u, v ,

则设 $w = u - v \quad \therefore \Delta w = \Delta u - \Delta v = 0$

则 w 在边界为 0, 根据调和函数性质, w 在区域内均为 0 即 $u = v$

f) 取一函数 v , 再任取一调和函数 u , 设 $w = v - u$

$$\begin{aligned} \text{则 } \int_{\Omega} |\nabla v|^2 dV \quad D(v) &= \int_{\Omega} |\nabla v|^2 dV = \int_{\Omega} |\nabla w + \nabla u|^2 dV \\ &= \int_{\Omega} |\nabla u|^2 dV + 2 \int_{\Omega} \nabla w \cdot \nabla u dV + \int_{\Omega} |\nabla w|^2 dV \end{aligned}$$

由 (1b) 有 $\int_{\Omega} \nabla u \cdot \nabla w dV + \int_{\Omega} w \Delta u dV = \int_{\partial\Omega} w \frac{\partial u}{\partial n} d\sigma$

而 $\Delta u = 0, \quad \partial\Omega \perp w = 0$ 故 $\int_{\Omega} \nabla u \cdot \nabla w dV = 0$

$\therefore D(v) = D(u) + D(w) \geq D(u)$

等号成立时 $w = 0, \quad v = u,$

15. c) $4\pi g(p) = \int_S (f \frac{\partial g}{\partial n} - g \frac{\partial f}{\partial n}) d\sigma = \frac{1}{R} \int_V \Delta g dV + \frac{1}{R^2} \int_S g d\sigma = \frac{1}{R^2} \int_S g d\sigma$
 $\therefore g(p) = \frac{1}{4\pi R^2} \int_S g d\sigma$

d) c) $4\pi g(p) = \int_S (g \nabla \frac{1}{r_p} - \frac{1}{r_p} \nabla g) d\sigma = \frac{1}{R^2} \int_S g d\sigma - \frac{1}{R} \int_V \Delta g dV = \frac{1}{R^2} \int_S g d\sigma$

故 $g(p) = \frac{1}{4\pi R^2} \int_S g d\sigma$

d) $\int_S \int_V g dV = \int_0^R \int_S g d\sigma dr = \int_0^R 4\pi r^2 g(p) dr = g(p) \frac{4\pi R^3}{3} = g(p) V(B)$

$\therefore g(p) = \frac{1}{V(B)} \int_B g dV$

16 同 13.4

17. 对点泰勒展开, $U(x_0) = U(x_0) + \nabla U(x_0) \cdot (x - x_0) + \frac{1}{2} (x - x_0)^T H (x - x_0) + O(|x - x_0|^3)$

即有 设点稳定平衡, 有 $\nabla U(x_0) = \nabla F(x_0) = 0$

则 $U(x) - U(x_0) = \frac{1}{2} (x - x_0)^T H (x - x_0) + O(|x - x_0|^3)$

$= \frac{1}{2} (x - x_0)^T (-\nabla F) (x - x_0) + O(|x - x_0|^3)$

由于该点稳定 $\nabla F = -\nabla U$ 故 $\nabla F(x_0) = -\nabla U(x_0) = -kI$

考虑 $\nabla F(x) = \nabla F(x_0) +$

$F(x) = F(x_0) + \nabla F(x_0)(x - x_0) + O(|x - x_0|^2) = \nabla F(x_0)(x - x_0) + O(|x - x_0|^2)$

当 $|x - x_0| \rightarrow 0$ 时, $F(x) = -k \nabla F(x_0)(x - x_0) = -k(x - x_0)$ (由于稳定) 这里 $k > 0$

$$\text{从而 } U(X) - U(X_0) = \frac{1}{2}(X-X_0)^T(K(X-X_0)) + O(|X-X_0|^3) = \frac{k}{2}|X-X_0|^2 + O(|X-X_0|^3) > 0$$

$\therefore U(X_0)$ 是一个极小点.

18. b) 对空心圆内任意两点 $A(r_1 \cos \theta_1, r_1 \sin \theta_1)$ 和 $B(r_2 \cos \theta_2, r_2 \sin \theta_2)$ 不妨设 $\theta_1 > \theta_2$

若 $\theta_1 - \theta_2 \in (0, \pi]$, 取路径为二者之间线段,

若 $\theta_1 - \theta_2 = 0$,

若 $\theta_1 - \theta_2 \in [\pi, 2\pi)$, 取路径为二者之间线段

若 $\theta_1 - \theta_2 = \pi$, 取一点 $C(r_1 \cos \theta_1, r_1 \sin \theta_1)$ 且满足 $\theta_2 \neq \theta_1, \theta_2$, 取路径为线段 AC, BC

综上, 空心圆连通

18. a) b) 取道路 $x^2 + y^2 = \frac{3}{4}$, 无法收缩到一个点, 故其不是单连通

d) 同 b), 不是单连通

c) 取道路 $x^2 + y^2 + z^2 = \frac{3}{4}$, 无法收缩到一个点, 故其不是单连通

19. a) 取 $w = \frac{x dy \wedge dz + y dz \wedge dx + z dx \wedge dy}{(x^2 + y^2 + z^2)^{3/2}}$

有 $dw = 0$

而若 w 是恰当的, 则有 η 使 $w = d\eta$

考虑 w 在 $S: x^2 + y^2 + z^2 = 1$ 上的积分, $\oint_S w = \oint_S (x dy \wedge dz + y dz \wedge dx + z dx \wedge dy) = 4\pi \neq 0$

矛盾故 w 是闭的但不恰当

b) 取 $F = \frac{(x, y, z)}{(x^2 + y^2 + z^2)^{3/2}}$

则 $\nabla \cdot F = \nabla \cdot \left(\frac{1}{r^3} \vec{r} \right) = 0$

而若 $F = \nabla \times A$, 考虑 $\iint_S F \cdot dS = \iint_S dS = 4\pi \neq 0$, 矛盾

故 F 无源但非旋度场

20. a) 若 A 满足 $B = \nabla \times A$, 则任意标量函数 φ , $A + \nabla \varphi$ 也满足

$$\nabla \times (A + \nabla \varphi) = \nabla \times A + \nabla \times \nabla \varphi = \nabla \times A = B$$

b) ① 设 $B = (0, 0, B_0)$

① $A = (0, B_0 x, 0)$

② $A = (-B_0 y, 0, 0)$

③ $A = (-p B_0 y, q B_0 x, 0)$ $p + q = 1$

④ $A = (-\frac{1}{2} B_0 y, \frac{1}{2} B_0 x, 0)$

c) 设 A' 满足 $\nabla \times A' = B$ 则 $\nabla \cdot A' = 0$ 且 $A - A' = \nabla \varphi$

$$\text{则 } \nabla \cdot A = \nabla \cdot (A' + \nabla \varphi) = \nabla \cdot A' + \nabla \cdot \nabla \varphi = 0$$

$$\therefore \Delta \varphi = -\nabla \cdot A'$$

$$d) -\frac{j}{\epsilon_0 c^2} = \nabla \times B = \nabla \times (\nabla \times A) = \nabla(\nabla \cdot A) - \nabla^2 A = -\nabla^2 A \quad \therefore \Delta A = \frac{j}{\epsilon_0 c^2}$$

21. 同13

22. 同6

23. a) $\text{div } v = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0$

$$\nabla \times v = (0, 0, \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}) = 0 \Leftrightarrow \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} = 0$$

b) 对 $(-v_y, v_x) = \text{grad } \psi$, 则 $\text{div grad } \psi = -\frac{\partial v_y}{\partial x} + \frac{\partial v_x}{\partial y} = 0$

故在局部存在 ψ

对 $(v_x, v_y) = \text{grad } \varphi$, 则 $\nabla \times \text{grad } \varphi = \frac{\partial v_y}{\partial x} + \frac{\partial v_x}{\partial y} = 0$

故在局部存在 φ

c) 考虑 $\nabla \varphi \cdot \nabla \psi = -v_x v_x v_y + v_x v_y = 0$

故等值线正交

流线应满足 $\frac{dx}{v_x} = \frac{dy}{v_y}$, 而在流线上 $d\psi = (-v_y)dx + (v_x)dy = (v_y dx - v_x dy) = 0$

故 $\psi = C$ 与个质微元的运动轨迹重合

d) $\nabla \Delta \varphi = \nabla \cdot (\nabla \varphi) = \nabla \cdot (-v_x, v_x) = 0 \quad \nabla \cdot (v_x, v_y) = 0$

$$\Delta \varphi = \nabla^2 \varphi$$

$$\Delta \psi = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = \frac{\partial}{\partial x}(-v_y) + \frac{\partial}{\partial y}(v_x) = -\frac{\partial v_y}{\partial x} + \frac{\partial v_x}{\partial y} = 0$$

故 φ 为调和函数

而 $\frac{\partial \varphi}{\partial x} = -\frac{\partial \psi}{\partial y} = v_x \quad \frac{\partial \varphi}{\partial y} = \frac{\partial \psi}{\partial x} = v_y$

e) $\frac{\partial}{\partial z} (\varphi + i\psi)(x+iy) = \frac{\partial}{\partial z} [(\varphi + i\psi)(x+iy) + i\psi x + \varphi y - \psi x - i\varphi y]$

$$= \frac{\partial}{\partial z} [$$

$$(\varphi + i\psi)x, f(z+\Delta z) - f(z) = \frac{\partial \varphi}{\partial x} \Delta x + \frac{\partial \varphi}{\partial y} \Delta y + i(\frac{\partial \psi}{\partial x} \Delta x + \frac{\partial \psi}{\partial y} \Delta y) + o(|\Delta z|^2)$$

$$= (\frac{\partial \varphi}{\partial x} + i\frac{\partial \psi}{\partial x}) \Delta x + i(\frac{\partial \varphi}{\partial x} + i\frac{\partial \psi}{\partial x}) \Delta y + o(|\Delta z|)$$

$$= (\frac{\partial \varphi}{\partial x} + i\frac{\partial \psi}{\partial x})(\Delta x + i\Delta y) + o(|\Delta z|)$$

$$\therefore \frac{f(z+\Delta z) - f(z)}{\Delta z} = \frac{\partial \varphi}{\partial x} + i\frac{\partial \psi}{\partial x} + o(|\Delta z|)$$

24. a) 这是直接得出的

b) $\nabla \times E = -\frac{\partial B}{\partial t} = -\frac{\partial}{\partial t}(\nabla \times A) = -\nabla \times (\frac{\partial A}{\partial t})$

从而 $\nabla \times (E + \frac{\partial A}{\partial t}) = 0$

故存在 $-\nabla \varphi = E + \frac{\partial A}{\partial t} \quad E = -\nabla \varphi - \frac{\partial A}{\partial t}$

c) $\nabla \times (A + \nabla \psi) = \nabla \times A + \nabla \times \nabla \psi = B$

$$-\nabla(\varphi - \frac{\partial \psi}{\partial t}) - \frac{\partial}{\partial t}(A + \nabla \psi) = -\nabla \varphi + \nabla \frac{\partial \psi}{\partial t} - \frac{\partial}{\partial t} A - \frac{\partial \psi}{\partial t} = -\nabla \psi - \frac{\partial}{\partial t} A = E$$

d) $\nabla \cdot E = \nabla \cdot (-\nabla \varphi - \frac{\partial A}{\partial t}) = -\nabla^2 \varphi - \frac{\partial}{\partial t} \nabla \cdot A = \frac{\rho}{\epsilon_0}$

e) $c^2 \nabla \times B - \frac{\partial E}{\partial t} = c^2 \nabla \times \nabla \times A - \frac{\partial}{\partial t}(A + \nabla \psi - \nabla \varphi - \frac{\partial A}{\partial t}) = -c^2 \nabla^2 A + c^2 \nabla(\nabla \cdot A) + \frac{\partial}{\partial t} \nabla \varphi + \frac{\partial^2 A}{\partial t^2} = \frac{j}{\epsilon_0}$

f) 设现在 $\tilde{\varphi}, \tilde{A}$ 满足 $B = \nabla \times \tilde{A}$ $E = -\nabla \tilde{\varphi} - \frac{\partial \tilde{A}}{\partial t}$

考虑 $A = \tilde{A} + \nabla \psi$ $\varphi = \tilde{\varphi} - \frac{\partial \psi}{\partial t}$ 使

$$\nabla \cdot A = -\frac{1}{c^2} \frac{\partial \varphi}{\partial t} \quad \text{即} \quad \nabla \cdot \tilde{A} + \Delta \psi = -\frac{1}{c^2} \frac{\partial \tilde{\varphi}}{\partial t} + \frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2}$$

$$\text{即} \quad \Delta \psi + \left(\pm + \frac{1}{c^2} \frac{\partial \tilde{\varphi}}{\partial t} + \nabla \cdot \tilde{A} \right) = \frac{1}{c^2} \frac{\partial^2 \psi}{\partial t^2}$$

根据题意该方程有解, 故 A, φ 存在

$$g) \quad -\nabla^2 \varphi - \frac{\partial}{\partial t} \nabla \cdot A = -\nabla^2 \varphi + \frac{\partial}{\partial t} \frac{1}{c^2} \frac{\partial \varphi}{\partial t} = -\nabla^2 \varphi + \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = \frac{\rho}{\epsilon_0}$$

$$\therefore \frac{\partial^2 \varphi}{\partial t^2} = c^2 \Delta \varphi + \frac{\rho c^2}{\epsilon_0}$$

$$\begin{aligned} \frac{j}{\epsilon_0} &= \frac{\partial^2 A}{\partial t^2} + \frac{\partial}{\partial t} \nabla \varphi + c^2 \nabla (\nabla \cdot A) - c^2 \nabla^2 A = \frac{\partial^2 A}{\partial t^2} + \frac{\partial}{\partial t} \nabla \varphi + c^2 \nabla \frac{\partial \varphi}{\partial t} - c^2 \nabla^2 A \\ &= \frac{\partial^2 A}{\partial t^2} - c^2 \nabla^2 A \end{aligned}$$

$$\therefore \frac{\partial^2 A}{\partial t^2} = \frac{j}{\epsilon_0} + c^2 \Delta A$$